Botany Labs

Alveary

Botany Labs

GRADE(S): 9-12

About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Botany

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Botany Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Botany				
Botany Lessons Nature Walks & Scouting: Grades 9-12	Botany Lessons	Botany Lessons	Botany Lessons Nature Notebook: Grades 9-12	Botany Lessons Botany Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- ∞ View Basic Supplies
- ∞ View Supply List Details

Botany Labs

Slides and Coverslips

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

Botany Labs

How To Approach Labs

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Botany Labs

[Click THIS text or scan the QR code for links.](#)

Term 1

WEEK 1

⬚ **60m Botany Labs - Lesson 1**

Materials: Elementary Studies in Plant Life

Prep: Read Student Tip.

→ OBSERVE
Notice a variety of plants this week, doing the practical work from Ch.1. Notice root and shoot, and how the leaves and branches are attached in the different cases. Notice plants in which flowers are fading, and fruit is ripening in their place. Some activities may be delayed depending on the time of year and your geographic location. Do what you can to gain the most from the chapter. If the plants or trees mentioned are unavailable locally, look up the item online. Also, you may need to visit local nurseries, botanical gardens, and neighbors' yards and gardens to find specimens to examine.

• STUDENT TIP
Labs and prompts for your field work are included in your lesson plans, but it is important to establish a personal habit of outdoor walks as a part of life!

WEEK 2

⬚ **60m Botany Labs - Lesson 2**

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.2. Record your work in your notebook.

WEEK 3

⬚ **60m Botany Labs - Lesson 3**

Materials: Link, household items indicated, microscope, and glass slides

Prep: Read Student Tip.

→ LAB DAY
Use the procedure at the link to evaluate microbial life in soil from at least 3 locations. The locations may be at the same site, but they should be some distance apart. Soil that is even 10-20 yards apart may be quite different. It is okay to be unsure of what you are viewing under the microscope.

→ VIEW
∞ Link: Soil Sampling Procedure

• STUDENT TIP
Plan to go to another type of habitat next week. If you collected soil from a neighborhood this week, perhaps go to a forest or a pond next week.